Unit 4 FRQ Practice
AP Calculus AB Name_____

NO CALCULATOR IS ALLOWED FOR THIS QUESTION

Ayan is driving due east and passes through an intersection at time $t = 0$ seconds. The velocity of Ayan's car can be modeled by the equation $v(t) = -\frac{1}{10}t^3 + \frac{3}{2}t^2 - 4t + 10$, for $0 \leq t \leq 11$, where $v(t)$ is measured in meters per second and t is measured in seconds. Ayan's car is 900 meters east of the intersection after 10 seconds.

(a) Find the acceleration of Ayan's car at time $t = 10$. Show the work that leads to your answer and indicate units of measure.

(b) Is Ayan's car speeding up or slowing down at time $t = 10$? Explain your reasoning.

(c) Ben is in a hot air balloon directly over the intersection. Let θ be the angle of elevation from Ayan to Ben, where θ is measured in radians. Let A be the distance between Ayan's car and the intersection and B be the distance between Ben and the intersection at time t, both measured in meters. These quantities satisfy the relationship $\tan\theta = \frac{B}{A}$. Write an equation that describes the relationship between the rates of change of θ, A, and B.

(d) Using θ as defined in Part C, at time $t = 10$, $\frac{d\theta}{dt} = -0.0073$. Using correct units, interpret the meaning of $\frac{d\theta}{dt}$ in the context of this problem.

(e) Keng is standing 1200 meters due north of the intersection. Find the rate of change of the distance between Keng and Ayan, in meters per second, at time $t = 10$. Show the work leading to your answer.